PHYSICS 1B · MIDTERM 2 CHEATSHEET

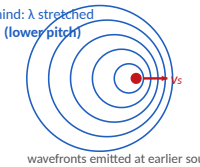
① Doppler · beats · sound ② H-atom orbit · escape · 2D ③ Van de Graaff sphere ④ Gauss · cube flux ⑤ Dipole ★ Other ch.4–7 Σ Constants · math

5 Qs · 100 min · calculator OK. Write the principle first (Gauss / energy cons. / F = mv²/r / superposition), **label middle steps** (rubric), draw the diagram with arrows, keep symbols until the end, then plug in numbers **in SI**.
Check: units, signs (+/− charge), limits (z≫d → point charge or kp/z³).

① DOPPLER EFFECT · BEATS

review p.1 · lect. 7

behind: λ stretched, **f ↓ (lower pitch)** — ahead: λ squeezed, **f ↑ (higher pitch)**



wavefronts emitted at earlier source positions

His review-slide form: put an x-axis pointing **from source → observer**; v_s, v_o = signed velocity components along it, **relative to the ground (still air)**:

$$f' = \frac{v - v_o}{v - v_s} f$$

v = speed of sound (343 m/s at 20 °C)

Source toward observer: $v_s > 0 \Rightarrow$ f'↑. Observer toward source: $v_o < 0 \Rightarrow$ f'↑.

Memory form $f' = f\,\frac{v \pm v_o}{v \mp v_s}$ with magnitudes: **upper sign = moving TOWARD** the other, lower = away. "Toward ⇒ f' higher" always.

case	λ'	f'
source approaching	$\frac{v-v_s}{f}$ squeezed	$\frac{v}{v-v_s} f$ higher
source receding	$\frac{v+v_s}{f}$ stretched	$\frac{v}{v+v_s} f$ reduced
observer approaching	λ unchanged	$\frac{v+v_o}{v} f$
observer receding	λ unchanged	$\frac{v-v_o}{v} f$

Moving source changes λ (wavefronts bunch up). A moving observer changes how fast wavefronts are met, so the relative speed changes, not λ.

Slow speeds ($v_s, v_o \ll v$): $\frac{\Delta f}{f} \approx -\frac{v_r}{v}$, $v_r = v_o - v_s$ (positive = separating).

His 3 follow-up questions

- **Car moving away?** f' is **reduced**: $f' = \frac{v}{v+v_s} f$.
- **Reflected by a wall?** A stationary wall **keeps** the frequency it receives. **Moving reflector** (speed u toward a stationary source/listener) = double Doppler: the wall hears $f_1 = f\frac{v+u}{v}$, then re-emits as a moving source: $f_2 = f_1 \frac{v}{v-u} = f\frac{v+u}{v-u}$. Beats with the original: $|f_2 - f|$.
- **Wind?** The speed of sound is modified: relative to the ground it travels at $v \pm w$ (downwind +). Work in the air frame (velocities relative to the air, v unchanged), or use $f' = \frac{v+w-v_o}{v+w-v_s} f$ with w along the source→observer axis. Source and observer both at rest ⇒ **no shift** (f' = f); only λ and travel time change.

Beats (at a fixed position): $f_{beat} = |f_1 - f_2|$

$y = 2A\cos\left(2\pi\frac{f_1-f_2}{2}t\right)\cos(2\pi f_{avg}t)$, $f_{avg} = \frac{f_1+f_2}{2}$. You hear pitch f_{avg} with loudness pulsing $|f_1 - f_2|$ times/s (e.g. 500 & 505 Hz give 5 beats/s, **not** a 5 Hz tone). The envelope has frequency Δf/2, but loudness peaks twice per envelope cycle, so beats = Δf. Piano tuner: 440 vs 442 gives 2 beats/s, and beats slow as the notes match.

Doppler + beats combo: e.g. a moving source with a stationary one at the same f: $f_{beat} = |f' - f|$. Two listeners or a reflector: find each f' separately, then subtract.

Light: receding gives $\Delta\lambda = \lambda_{obs} - \lambda_{src} > 0$ (**redshift**); approaching gives blueshift (Andromeda). Hubble $v = H_0 d$, age $\approx 1/H_0 \approx 14$ Gyr. Standard candle $F = \frac{L}{4\pi d^2} \Rightarrow d = \sqrt{\frac{L}{4\pi F}}$. Binary stars and exoplanets: periodic shifts.

Shock wave: Mach $M = \frac{v_{wf}}{c_s}$. Cone: $\sin\theta = \frac{c_s t}{v_{src} t} = \frac{1}{M}$ (M = 2 gives θ = 30°). The boom is heard whenever the cone passes you, continuously, not only when "breaking the barrier".

① cont. · SOUND · INTENSITY · PIPES

lect. 7

Sound is a longitudinal pressure wave in air and fluids (no shear); solids carry both kinds. $s = s_{max}\cos(kx \mp \omega t + \varphi)$, $\Delta P = \Delta P_{max}\sin(kx \mp \omega t + \varphi)$: 90° out of phase with s. Hearing 20 Hz–20 kHz.

Speed $= \sqrt{\frac{\text{restoring}}{\text{inertia}}}$: fluid $\sqrt{B/\rho}$, solid $\sqrt{Y/\rho}$, ideal gas $v = \sqrt{\gamma RT_K/M}$ (adiabatic pV^γ=const gives $dp/d\rho = \gamma p/\rho$). Air: 331 m/s at 0 °C, 343 at 20 °C, $v \approx 331\sqrt{T_K/273}$. $B = -\frac{\Delta p}{\Delta V/V}$, $Y = \frac{F/A}{\Delta L/L}$. Faster in solids. Sound bends toward the slower medium (SOFAR channel).

Intensity $I = \frac{P}{4\pi r^2}$ (2r gives 1/4, −6 dB).

$I = \frac{\langle\Delta p_{max}\rangle^2}{2\rho v} = \frac{1}{2}\Delta p_{max} u_{max}$, $u_{max} = \frac{\Delta p_{max}}{\rho v} = \omega s_{max}$. $I \propto A^2$.

Level $\beta = 10\log_{10}\frac{I}{I_0}$ dB, $I_0 = 10^{-12}$ W/m² (threshold), pain ≈ 1 W/m² ≈ 120 dB. 10×I adds 10 dB; 2×I adds 3 dB; N equal sources add $10\log N$. $I = I_0 10^{\beta/10}$. ~1 dB is just noticeable; 3 dB is easily noticed. Loudness (phon) also depends on f.

Interference (two speakers in phase): $\Delta r = |r_2 - r_1| = n\lambda$ is **constructive**; $\Delta r = (n + \frac{1}{2})\lambda$ is **destructive**. Phase φ = 2πΔr/λ, $A_R = 2A|\cos\frac{\varphi}{2}|$. Noise cancelling: anti-sound at 180°.

OPEN–OPEN (all n) — CLOSED–OPEN (odd n)



displacement antinode at open end, node at closed end

Open–open: $\lambda_n = \frac{2L}{n}$, $f_n = \frac{nv}{2L}$, **n = 1, 2, 3...** (slide says "1, 3, 5", but its own picture shows λ = 2L, L, 2L/3: all n).

Closed–open: $\lambda_n = \frac{4L}{n}$, $f_n = \frac{nv}{4L}$, **odd n only**. String fixed–fixed: $f_n = \frac{n}{2L}\sqrt{T_T/\mu}$. Vocal folds $f \sim \frac{1}{2L}\sqrt{T_T/\mu}$ (shorter and lighter give higher f). Timbre = the mix of harmonics $f_1, 2f_1, 3f_1 \ldots$; resonance means driving at a natural frequency.

② H-ATOM: ORBIT · ESCAPE · 3D vs 2D

review p.2

Proton +e, electron −e, $r = 5.29\times10^{-11}$ m. **Electric force is the centripetal force**; ignore gravity.

$F = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2} = \frac{(8.99\times10^9)(1.602\times10^{-19})^2}{(5.29\times10^{-11})^2} = \mathbf{8.25\times10^{-8}\ N}$ (attractive). Gravity $F_G = G\frac{m_e m_p}{r^2} = 3.63\times10^{-47}$ N. Ratio ≈ 2×10³⁹.

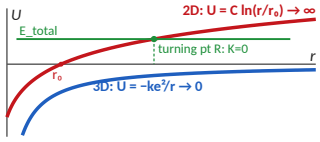
1. **Orbit speed**: $\frac{m_e v^2}{r} = \frac{ke^2}{r^2} \Rightarrow v = \sqrt{\frac{ke^2}{m_e r}} = \mathbf{2.19\times10^6}$ **m/s**. Period $T = \frac{2\pi r}{v} = 1.52\times10^{-16}$ s.
2. **Energies** (U(∞)=0): $U = -\frac{ke^2}{r} = -27.2$ eV, $K = \frac{1}{2}m_e v^2 = \frac{ke^2}{2r} = +13.6$ eV, $E = K + U = -\frac{ke^2}{2r} =$ **−13.6 eV** (bound, E<0). Ionization energy 13.6 eV.
3. **Escape**: reach ∞ with zero speed, $E_i = E_f = 0$:

$\frac{1}{2}m_e v_{esc}^2 - \frac{ke^2}{r} = 0 \Rightarrow v_{esc} = \sqrt{\frac{2ke^2}{m_e r}} = \sqrt{2}\,v_{orb} =$ **3.10×10⁶ m/s**. Same structure as gravity's $v_{esc} = \sqrt{2GM/r}$ (swap GMm for ke²). To escape you need **KE larger than |PE|**.

2D world: F ∝ 1/r (given: $F_r = -\frac{C}{r}$, C = "ke²" > 0). Why: field lines spread over a circumference $2\pi r$, not an area $4\pi r^2$ (same as an infinite line charge, $E = \frac{\lambda}{2\pi\epsilon_0 r}$).

1. $F = -\frac{dU}{dr} \Rightarrow \frac{dU}{dr} = \frac{C}{r}$. **Integrate**: $U(r) = C\ln r +$ const. Choose U(r₀) = 0 (any reference):
$U(r) = C\ln\frac{r}{r_0}$ (ln r − ln r₀ = ln(r/r₀))
2. Start at r₀ with v_0, stop at max separation R (K = 0):
$\frac{1}{2}m_e v_0^2 = U(R) - U(r_0) = C\ln\frac{R}{r_0}$
3. $v_0 = \sqrt{\frac{2C}{m_e}\ln\frac{R}{r_0}}$. As R→∞, $\ln(R/r_0) \to \infty$, so $v_{escape} \to \infty$. No finite escape velocity in 2D.

U grows without bound as r→∞ (in 3D U→0). Any finite energy has a turning point where K = E − U = 0. 2D circular orbit: $\frac{mv^2}{r} = \frac{C}{r} \Rightarrow v = \sqrt{C/m}$, the same at every r! C has different units from 3D k; use the model given.



any E > 0 escapes in 3D; 2D always turns back

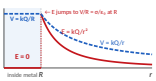
③ VAN DE GRAAFF · HOLLOW METAL SPHERE

review p.3

Belt carries charge up to a **hollow conducting dome** (radius R, charge Q). The prof confirmed: **metal sphere only**, not uniformly charged plastic. Charge sits on the **outer surface**.

Gauss: imagine a sphere of radius r: $E \cdot 4\pi r^2 = \frac{Q_{enc}}{\epsilon_0}$. r < R: $Q_{enc} = 0 \Rightarrow E = 0$. r > R: $E = \frac{kQ}{r^2}$, like a point charge.

Potential (V(∞)=0), $V = -\int_\infty^r E\,dr$:



r ≥ R: $V = \frac{kQ}{r}$

r ≤ R: $V = \frac{kQ}{R}$ = const ≠ 0

Surface: $E(R^+) = \frac{kQ}{R^2} = \frac{V}{R} = \frac{\sigma}{\epsilon_0}$

"E = 0 inside but V = 100 kV?" $E_r = -\frac{dV}{dr} = 0$ means V is **constant**, not zero. Its value is fixed by continuity with the outside at r = R. E depends on the **gradient** of V, not its value.

Lecture example: diameter 25.0 cm, so R = 0.125 m, V = 100 kV.

1. Charge: $Q = 4\pi\epsilon_0 RV = \frac{RV}{k} = \frac{(0.125)(10^5)}{8.99\times10^9} =$ **1.39 μC** (small Q, yet huge V)
2. Surface field: $E = \frac{V}{R} = \frac{10^5}{0.125} =$ **8.0×10⁵ V/m** < 3×10⁶ (no breakdown). σ = Q/4πR² = 7.08×10⁻⁶ C/m².
3. **Max charge before air breaks down** ($E_{bd} \approx 3\times10^6$ V/m): $\frac{kQ_{max}}{R^2} = E_{bd} \Rightarrow Q_{max} = \frac{E_{bd}R^2}{k} =$ **5.2 μC**, $V_{max} = E_{bd}R =$ **375 kV**.
4. **Work to bring q from ∞ to the surface** (slowly): $W_{ext} = q\,\Delta V = q\frac{kQ}{R}$ (field does $-qV$). Example: q = 1 μC at 100 kV gives 0.1 J. Released from the surface, +q gains $K = qV$ going to ∞.
5. Outside: at r = 2R, V = 50 kV, E = 2×10⁵ V/m; V(∞) = 0.

Scaling: fixed Q, 2R gives V/2, E/4. Fixed V, 2R gives 2Q, E/2.

Capacitance $C = 4\pi\epsilon_0 R = R/k = 13.9$ pF. Energy $U = \frac{1}{2}QV = \frac{1}{2}CV^2 =$ **0.0695 J** (slide prints 0.695 J, a typo).

Charge q placed inside the cavity (★): −q is induced on the inner wall and +q added to the outer surface, so E ≠ 0 in the cavity but still 0 in the metal. **Faraday cage**: an empty closed metal shell has E = 0 inside whatever the outside field (grounding not needed). **Plastic** uniformly charged sphere (★, not tested): inside $E = \frac{kQr}{R^3}$, $V = \frac{k(3R^2 - r^2)}{2R^3}$.

Sharp points: small sphere $E = \frac{V}{R}$. At fixed V, smaller R gives a bigger E (corona, lightning rods). Needle R = 0.1 mm: breakdown at $V = E_{bd}R = 300$ V.

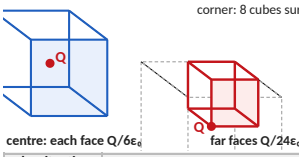
④ GAUSS'S LAW · CUBE FLUX

review p.4 · lect. 10

Flux (uniform E, flat surface): $\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta$, θ measured from the **normal** (not the plane!). General $\Phi = \int \vec{E}\cdot\hat{n}\,dA$. Closed surface: **normal points outward**. Out is +, in is −, parallel to the face is 0. Units N·m²/C.

Gauss: $\oint \vec{E}\cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$. The surface must be closed and orientable (a Klein bottle won't do). Q_{enc} is the algebraic sum inside. Outside charges change the local E on the surface but add **zero net flux**.

corner: 8 cubes surround Q



centre: each face Q/6ε₀ — **far faces Q/24ε₀ · near 0**

cube situation (side L)	result
uniform E along +x (no charge)	entry face $-EL^2$, exit $+EL^2$, 4 sides 0, **net 0**
Q at **centre**	total $\frac{Q}{\epsilon_0}$, **each face** $\frac{Q}{6\epsilon_0}$ (symmetry)
Q moved off-centre (still inside)	total **unchanged**; individual faces now **unequal**
Q at a **corner**	8 cubes would enclose Q, so this cube gets $\frac{Q}{8\epsilon_0}$. The 3 far faces get $\frac{Q}{24\epsilon_0}$ each; the 3 faces touching Q get 0 (E runs along them)
Q at a face centre / edge midpoint	2 cubes give $\frac{Q}{2\epsilon_0}$ / 4 cubes give $\frac{Q}{4\epsilon_0}$
Q **outside** the cube	faces ≠ 0 but **net 0** (what enters leaves)
+5 μC and −8 μC inside	$\frac{-3\times10^{-6}}{8.85\times10^{-12}} = -3.39\times10^5$ N·m²/C
$\vec{E} = ax\hat{i} + by\hat{j}$ in $[0, L]^3$	x = L face: $aL\cdot L^2$, x = 0 face: 0, same for y, so $\Phi = (a+b)L^3$, $Q = \epsilon_0(a+b)L^3$

Pattern: charge at centre/face/edge/corner shares 1/1, 1/2, 1/4, 1/8 of Q/ϵ_0. **Why 0 on the near faces:** the field from a corner charge points radially, i.e. **along** the 3 planes that meet at the corner, so $\vec{E}\cdot\hat{n} = 0$.

Lecture examples: 10 N/C at 30° above the xy-plane, through 6 m² in the xz-plane (n = ŷ): $\Phi = EA\cos 30^\circ = 52$ N·m²/C. $\vec{E} = cy^2\hat{k}$ through an a×b rectangle in xy, strips dA = b dy: $\Phi = cb\int_0^a y^2 dy = \frac{1}{3}a^3bc$. Blobs: count only the charges inside, e.g. −4+6−1 = +1 μC.

Strategy + standard results

Symmetry → surface where E is constant and ∥ or ⊥ to n̂.

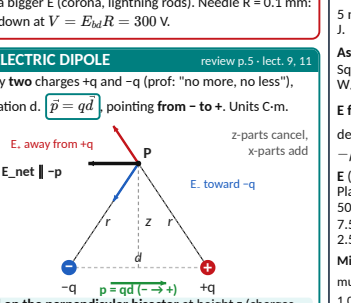
Sphere $E\,4\pi r^2$ · **cylinder** $E\,2\pi rL$ · **pillbox** $2EA$.

- Uniform insulating sphere ρ_0: in $E = \frac{kQr}{R^3}$ (∝ r), out $\frac{kQ}{r^2}$; max at R. $\rho = ar^n$: $Q_{enc} = \frac{4\pi a}{n+3}r^{n+3}$, in $E = \frac{ar^{n+1}}{(n+3)\epsilon_0}$, out $\frac{aR^{n+3}}{(n+3)\epsilon_0 r^2}$.
- Cylindrical shell σ_0: out $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{R\sigma_0}{\epsilon_0 r}$, in 0. Infinite plane $\frac{\sigma}{2\epsilon_0}$ (independent of distance). Two opposite planes: $\frac{\sigma}{\epsilon_0}$ between, 0 outside.
- **Conductor** (electrostatic equilibrium): E = 0 inside the metal and in an empty cavity (Faraday cage, no ground needed); excess charge on the outer surface; just outside $E = \frac{\sigma}{\epsilon_0}$, ⊥ to the surface; the whole conductor is equipotential.
- Differential form $\nabla\cdot\vec{E} = \frac{\rho}{\epsilon_0}$ (divergence theorem). Zero divergence ≠ zero E (a uniform field has div 0).

⑤ ELECTRIC DIPOLE

review p.5 · lect. 9, 11

Exactly **two** charges +q and −q (prof: "no more, no less"), separation d. $\vec{p} = q\vec{d}$, pointing **from − to +**. Units C·m.



Field on the perpendicular bisector at height z (charges at x = ±d/2):

1. Each charge: $|E_\pm| = \frac{kq}{r^2}$, $r = \sqrt{z^2 + (d/2)^2}$
2. Geometry: $\cos\theta = \frac{z}{r}$, $\sin\theta = \frac{d/2}{r}$ (θ from the z-axis)
3. z-components **cancel**; x-components add: $E = 2\frac{kq}{r^2}\sin\theta$
4. $E = \frac{kqd}{(z^2 + d^2/4)^{3/2}} = \frac{kp}{(z^2 + d^2/4)^{3/2}}$, direction **opposite p** (parallel to the line from + to −)
5. z ≫ d: $E \approx \frac{kp}{z^3}$, falls off faster than a point charge (net charge 0)

Compare: two equal +q: x-parts cancel, $E_z = \frac{2kqz}{r^3} \to \frac{k(2q)}{z^2}$ (point charge 2q). **On the axis** (distance z from the centre): $E = \frac{2kpz}{(z^2 - d^2/4)^2} \to \frac{2kp}{z^3}$, along p.

Potential (scalar, no components!): $V = kq\left(\frac{1}{r_+} - \frac{1}{r_-}\right)$, $r_\pm = \sqrt{x^2 + (z \mp \frac{d}{2})^2}$. Far: $V = \frac{k\vec{p}\cdot\hat{r}}{r^2} = \frac{kp\cos\theta}{r^2}$. On the bisector: **V = 0 but E ≠ 0.**

Lecture: q = 3 nC, d = 4 cm: V(0,0,1 cm) = 1.80×10³ V; V(0,0,−5 cm) = −514 V; V(3,0,2 cm) = 360 V.

Dipole in a uniform external E: $\vec{F}_+ = q\vec{E}$, $\vec{F}_- = -q\vec{E}$, so **net force 0**, but the forces are separated, giving a **torque**: $\vec{\tau} = \vec{\frac{d}{2}} \times q\vec{E} + (-\vec{\frac{d}{2}}) \times (-q\vec{E}) = \vec{p}\times\vec{E}$, $|\tau| = pE\sin\theta$

Energy: $dU = -dW = pE\sin\theta\,d\theta \Rightarrow$ $U = -\vec{p}\cdot\vec{E} = -pE\cos\theta$ (U = 0 at 90°). $\tau_\theta = -\frac{dU}{d\theta}$ (like $F = -\frac{dU}{dx}$).

θ	U	τ	
0°	−pE (min)	0	stable, aligned
90°	0	pE (max)	
180°	+pE (max)	0	unstable

Work by an external agent to rotate $\theta_1 \to \theta_2$: $W = U_2 - U_1 = pE(\cos\theta_1 - \cos\theta_2)$. Small oscillations: $I\ddot{\theta} = -pE\theta \Rightarrow \omega = \sqrt{pE/I}$.

Example: q = 2 μC, d = 0.10 m, so p = 2×10⁻⁷ C·m. E = 5×10⁴ N/C, θ = 30°; τ = pE sin30° = **5.0×10⁻³ N·m**, U = −pE cos30° = −8.7×10⁻³ J.

Shifted / asymmetric charges: never assume cancellation. For each charge: $\vec{r} = (\text{P} - \text{source})$, $E_x = \frac{kq_i\Delta x}{r^3}$, $E_y = \frac{kq_i\Delta y}{r^3}$ (sign of q included), then add components: $|E| = \sqrt{E_x^2 + E_y^2}$, $\theta = \tan^{-1}\frac{E_y}{E_x}$.

Non-uniform E also gives a net force. Induced dipole: an atom polarizes in E. Water is a permanent dipole. Microwaves: lagging polarization absorbs $\langle p\rangle = \frac{1}{2}\omega E_0 P_0 \sin\delta$. Flocking: $q\vec{E}$ moves fibres, $\vec{p}\times\vec{E}$ stands them up.

★ CHARGE · COULOMB · FIELDS

lect. 8–9

Charge is **quantized** (q = ne, e = 1.602×10⁻¹⁹ C; quarks ±⅓e, ±⅔e) and **conserved**, locally too: $\frac{dQ_{in}}{dt} = -I_{out}$. Conductors have mobile electrons; insulators have bound ones. A charged comb attracts neutral paper by **polarization** (near charges closer). **Induction**: rod near a sphere → ground it → cut the ground → remove the rod: the sphere keeps the opposite charge.

Coulomb $\vec{F}_{12} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}$ (on 1 from 2; like charges repel). Newton's 3rd law holds. Both Coulomb and gravity go as 1/r² because "flux" spreads over 4πr² (3D). Superposition: add vectors.

Ex: $q_2 = -3e$ at the origin, $q_1 = +2e$ at (0, d), $q_3 = -5e$ at (2d, 0): $F_{21} = \frac{k2e^2}{d^2}$ (+y), $F_{23} = \frac{k15e^2}{4d^2}$ (−x), $F = \sqrt{F_{21}^2 + F_{23}^2}$, $\theta = \tan^{-1}\frac{F_{21}}{F_{23}}$.

Field $\vec{E} = \vec{F}/q$, $\vec{E} = \sum\frac{kq_i}{r_i^2}\hat{r}_i$: away from +, toward −. He^+ example: E = k(2e)/r² = 4.1×10¹² N/C at 26.5 pm. Field lines start on + and end on −, number ∝ q, tangent = E, density = |E|, never cross.

source (on axis, distance z)	E	V
point q	$\frac{kq}{r^2}$	$\frac{kq}{r}$
line segment L (bisector)	$\frac{kL}{z\sqrt{z^2+L^2/4}}$	$k\lambda\ln\frac{L+\sqrt{L^2+4z^2}}{-L+\sqrt{L^2+4z^2}}$
infinite line λ	$\frac{\lambda}{2\pi\epsilon_0 z}$	$-\frac{\lambda}{2\pi\epsilon_0}\ln\frac{r}{r_R}$
ring R, q	$\frac{kqz}{(z^2+R^2)^{3/2}}$	$\frac{kq}{\sqrt{z^2+R^2}}$
disk R, σ	$\frac{\sigma}{2\epsilon_0}\left(1 - \frac{z}{\sqrt{z^2+R^2}}\right)$	$\frac{\sigma}{2\epsilon_0}\left(\sqrt{z^2+R^2} - \lvert z\rvert\right)$
infinite plane σ	$\frac{\sigma}{2\epsilon_0}$ const	$-\frac{\sigma}{2\epsilon_0}\lvert z\rvert$+c

All reduce to kq/z^2 when z ≫ size. dq = λdl, σdA, ρdV; use symmetry to cancel components first.

★ POTENTIAL · ENERGY · APPLICATIONS

lect. 11

Conservative: $\oint\vec{E}\cdot d\vec{l} = 0$ (path-independent).

$W_e = -\Delta U$, $V = \frac{U}{q}$, $\Delta U = q\Delta V$,

$V_B - V_A = -\int_A^B \vec{E}\cdot d\vec{l}$. **Two-charge energy** $U = \frac{kQq}{r}$ (property of the system; >0 for like charges). Point $V = \frac{kq}{r}$; many charges $V = \sum\frac{kq_i}{r_i}$ (scalar sum).

Energy conservation: $K_i + qV_i = K_f + qV_f$. Through ΔV from rest: $\frac{1}{2}mv^2 = |q\Delta V|$ (electron through 100 V: 5.93×10⁶ m/s). 1 eV = 1.602×10⁻¹⁹ J. 3 nC pushed by a fixed 5 nC from 10 to 15 cm: $K = kQq(\frac{1}{r_1} - \frac{1}{r_2}) = 4.5\times10^{-7}$ J.

Assembly work $W = \sum_{i<j}\frac{kq_iq_j}{r_{ij}}$ (order doesn't matter). Square of 2, 3, 4, 5 μC with 1 cm sides: W_2 = 5.4, W_3 = 15.9, W_4 = 36.5, total 57.8 J.

E from V: $\vec{E} = -\nabla V$, $E_r = -\frac{dV}{dr}$. E points toward decreasing V; V = 0 does not mean E = 0. Poisson $\nabla^2 V = -\rho/\epsilon_0$. Equipotentials ⊥ E, and W = 0 along them. **Uniform E** (plates): $V_A - V_B = Ed$, $E = \frac{|\Delta V|}{d}$; 1 N/C = 1 V/m. Plates with σ = 6.81×10⁻⁷: E = σ/ε₀ = 7.69×10⁴ V/m, ΔV = 500 V at 6.5 mm. Max for 2.5 cm of air: 3×10⁶ × 0.025 = 7.5×10⁴ V. Battery: q = ΔU/ΔV; a 30 W lamp at 12 V passes 2.5 C/s = 1.56×10¹⁹ e/s.

Millikan: $|q|E = mg \Rightarrow |q| = \frac{mg}{E} = \frac{mgd}{V}$; charges come in multiples of e. **Inkjet**: $a_y = \frac{qE}{m}$, $t = \frac{L}{v_0}$, $y = \frac{qE}{2mv_0^2}x^2$ (example: 1.0×10³ m/s², 1 ms, 0.50 mm). **Precipitator**: $|q|E = 6\pi\eta r v_d \Rightarrow v_d = \frac{|q|E}{6\pi\eta r}$. **Photocopier**: light discharges the drum; toner sticks to charged (CAD) or exposed (DAD) areas. Spray paint wraps around; Faraday recesses are poorly coated.

✔ 5-QUESTION PLAYBOOK

review Qs

Q	likely asks	do this
①	f' approaching/receding; wall echo; beats; wind	Pick the axis source→observer, signs, $f' = \frac{v-v_o}{v-v_s}f$. Reflector = 2 steps. Beats = $\lvert f_1 - f_2\rvert$. Wind changes v only.
②	F on e⁻; orbit v; v_esc; 2D version	Coulomb; $\frac{mv^2}{r} = \frac{ke^2}{r^2}$; $E_{tot} = 0$ at ∞; 2D: $F = -\frac{dU}{dr}$, integrate to $C\ln\frac{r}{r_0}$, so v_esc → ∞.
③	E(r), V at surface, work ∞→R, max Q	Gaussian sphere: 0 / kQ/r². V = kQ/R. W = qV. $Q_{max} = E_{bd}R^2/k$.
④	flux total/per face: centre, moved, corner	Q_{enc}/ϵ_0; centre ÷6; moved: same total; corner ÷8, far faces ÷24, near 0.
⑤	V, E at P, z≫d, torque	V scalar sum; E by components (bisector $\frac{kp}{r^3}$, → kp/z³); $\tau = pE\sin\theta$, $U = -pE\cos\theta$.

Sign sanity: +q moving to lower V gains K. An electron (q < 0) moving to higher V gains K. E points from high V to low V. F on −q is opposite E.

Σ CONSTANTS · MATH

calculator allowed

$k = \frac{1}{4\pi\epsilon_0} = 8.99\times10^9$ N·m²/C² · $\epsilon_0 = 8.85\times10^{-12}$ C²/N·m² · $1/\epsilon_0 = 1.13\times10^{11}$ · e = 1.602×10⁻¹⁹ C · $m_e = 9.11\times10^{-31}$ kg · $m_p = 1.673\times10^{-27}$ kg · G = 6.67×10⁻¹¹ · $E_{bd,air} \approx 3\times10^6$ V/m · v_sound 343 m/s (20 °C) · R = 8.31 J/mol·K · $I_0 = 10^{-12}$ W/m² · μ = 10⁻⁶, n = 10⁻⁹, p = 10⁻¹².

$\int\frac{dx}{(z^2+x^2)^{3/2}} = \frac{x}{z^2\sqrt{z^2+x^2}}$ (x = z tanθ) · $\int\frac{dy}{\sqrt{x^2+y^2}} = \ln(y + \sqrt{x^2+y^2})$ · $\int\frac{dr}{r^2} = -\frac{1}{r}$ · $\int\frac{dr}{r} = \ln r$ · $\ln a - \ln b = \ln\frac{a}{b}$ · $\log_{10}2 = 0.301$, $\log_{10}3 = 0.477$ · $(1+x)^n \approx 1 + nx$ · sphere $4\pi r^2$, $\frac{4}{3}\pi r^3$; cylinder side $2\pi rL$ · $\sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$, $\cos a + \cos b = 2\cos\frac{a+b}{2}\cos\frac{a-b}{2}$.